

Universal Intelligence System

Australian computer scientist David Dowe of Monash University and Spanish researcher José Hernández-Orallo of the Polytechnic University of Valencia came together to devise a new system to measure intelligence itself. Their work is based on a theorem called the “Kolmogorov complexity”, which basically states that any object’s complexity is a measure of the computational resources needed to express it. And the minimal resources needed in doing so is the Kolmogorov complexity of that object.

To put it in layman’s terms, the amount of data needed to describe an object is how complex it is. Of course, each object has different ways by which you can describe it. A “red book” can also be called a “collection of papers bound together in a casing whose appearance is closest to the longest wavelength of light as can be seen by a human eye”. In this example, the object’s Kolmogorov complexity would be “red book”.

Of course, for testing intelligence, you would use different objects like strings of characters or binary. And that’s what Dowe and Hernández-Orallo are doing. By having a quantifiable measure for intelligence in the form of the Kolmogorov complexity, they’re able to judge whether one AI system is more intelligent than another. 